Credit Name: CSE 2140 2nd Language Programming

Assignment Name: printing.java

The following questions can help you in thinking critically about your problem-solving processes:

Understanding the Problem

How did you approach understanding the challenge?

I reviewed the printing price tier table in the textbook and the expected output example to see how quantity thresholds dictate different price brackets per copy.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

Making sure the boundary conditions for the if-else if ranges didn't overlap or leave gaps was tricky at first. I resolved it by carefully checking the numeric limits for each tier (e.g., 0-99, 100-499, etc.)

Planning the Solution

Did you create a plan or break the problem into smaller steps before coding?

Yes, I broke it down into taking user input, evaluating the tiered pricing structure using conditional statements, performing the final multiplication, and formatting the output strings

How did you decide on the tools, data structures, or algorithms to use?

I used int for copy counts since you can't print a partial copy, and double for prices and totals to accurately handle decimal currency values.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

I wrote it in pieces: first setting up the Scanner and variables, then building the if-else if ladder, and finally adding the cost calculation and print statements.

How did you test your solution along the way to make sure it was working?

I tested it using the sample inputs from the textbook screen dump (1001 and 678) to verify my outputs matched the correct pricing tiers.

Overcoming Challenges

What part of the problem was the most difficult for you?

Structuring the sequential order of the if-else if statements so that higher thresholds like > 1000 or ranges were evaluated correctly without triggering earlier conditions prematurely.

How did you handle moments when you felt stuck or unsure of what to do next?

I traced through the logic manually with sample numbers to see which condition branch would trigger first.

Learning

Was there anything you learned that you think will help you with future challenges?

Was there anything you learned that you think will help you with future challenges? I got a lot more comfortable using multi-branch if-else if structures to handle tiered pricing schedules, which will be useful for billing or discount programs.